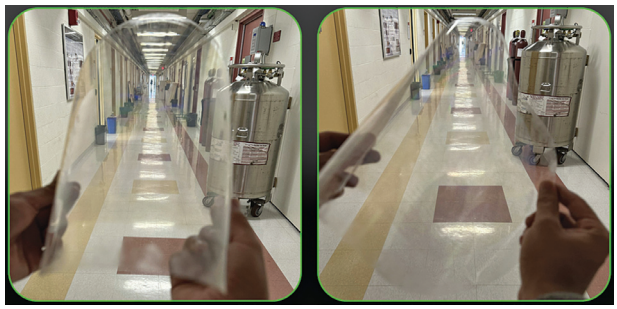
Flexible photonics reshape chip electronics

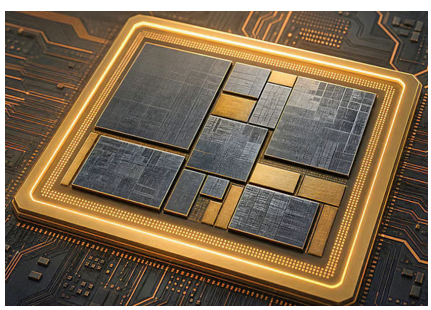
MIT researchers have developed a wafer-scale fabrication platform that makes silicon-photonics chips mechanically flexible and optically transparent without abandoning established semiconductor manufacturing workflows. Fabricated on 300mm wafers, the technology enables scalable production of photonic integrated circuits for curved and transparent systems. Demonstrated devices maintained performance after being repeatedly bent thousands of times around narrow cylinders, while the transparent finished architectures exhibited limited haze and distortion. The infrastructure could enable body-conforming health-monitoring systems, flexible optical sensors, and transparent augmented-reality displays integrated into protective visors, moving integrated photonics beyond conventional flat-chip architectures.



Photonics beyond conventional flat-chip architectures (Credit: https://news.mit.edu)

Adaptive chips can now connect to chiplets

AMD will offer the first Versal adaptive system-on-chip (SoC) with native support for UCIe 1.1 interfaces, enabling low-power, high-bandwidth communication with co-packaged specialised chiplets. The Versal RF series will be the first to support the technology, accommodating up to four UCIe-SP and two UCIe-AP interfaces to deliver multi-terabit-per-second aggregate bandwidth. The architecture allows individual functions to be upgraded or replaced without redesigning the entire SoC. By moving board-level serialised connections inside a single package, it reduces latency, size, development cost, and power consumption. The approach targets flexible, scalable systems for AI, RF, networking, and communications applications.



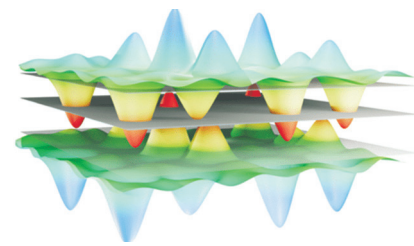
AMD to bring UCIe connectivity to select AMD Versal Adaptive SoCs (Credit: https://newsroom.amd.com/news)

Magnetic modem enables Arctic communication

MIT Lincoln Laboratory researchers have demonstrated magneto-inductive communication through Arctic sea ice, offering a potential way to connect distributed underwater sensors without conventional radio signals. Developed with Norway-based Havguard, the prototype uses changing magnetic fields, which can travel through seawater more effectively than RF signals. During Arctic testing, a remotely operated vehicle deployed the transmitter beneath 1.1 metre (3.6 feet) of ice, achieving alpha-system data rates of about 1.2 kilobytes per second. The technology could support remote monitoring of ice conditions, marine activity, and underwater environments while reducing the need for frequent human access.

Scandium nitride films break the Boltzmann thermopower limit

Researchers from JNCASR and IISc have demonstrated a colossal thermoelectric response in crystalline semiconductors, opening new possibilities for thermoelectrics and ultrasensitive temperature sensing. Heavily doped, highly compensated epitaxial scandium nitride (ScN) thin films achieved a Seebeck coefficient exceeding $-124 mV K^{-1}$ near room temperature, well above conventional limits. Randomly distributed charged dopants generate potential fluctuations that distort electronic bands, giving rise to percolative transport. In ultrathin films, modifications of the Coulomb potential further amplify these fluctuations to enhance the Seebeck response, enabling future applications in quantum-precision metrology.



Randomly distributed dopants create potential fluctuations that force charge carriers to move through percolative transport (Source: JNCASR/DST)

Hybrid energy storage reduces stress on EV batteries

NIT Rourkela has patented a hybrid energy storage system that reduces stress on electric vehicle (EV) batteries across 24V to 60V DC low-voltage platforms. Developed by Monalisa Pattnaik, Pradyumna Kumar Behera, and Karan Gupta, the architecture pairs a battery pack with a supercapacitor using a single converter, inductor, and control loop. The supercapacitor absorbs and supplies transient power during sudden acceleration and regenerative braking. Performance testing maintained a stable 48V output, ensuring smoother current transitions for electric scooters, rickshaws, and industrial utility vehicles.